

Where Prices Come From: The Interaction of Supply and Demand

ECON 101 - Macroeconomics

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- 3.1 The Demand Side of the Market
- 3.2 The Supply Side of the Market
- 3.3 Market Equilibrium: Putting Buyers and Sellers Together
- 3.4 The Effect of Demand and Supply Shifts on Equilibrium

3.1 Learning Objective

Goal

Discuss the variables that influence demand.

- We begin by investigating how **buyers** behave.
- **Market demand:** the demand by all the consumers of a given good or service.
- Key tools:
 - Demand schedule
 - Demand curve

Figure 3.1 – A Demand Schedule and Demand Curve (1 of 3)

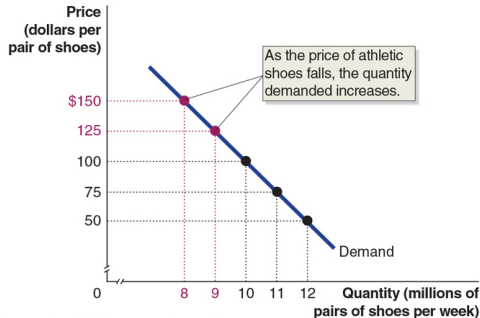
- **Demand schedule:** a table that shows the relationship between the **price** of a product and the **quantity of the product demanded**.
- **Demand curve:** a curve that shows the relationship between the price of a product and the quantity of the product demanded.

Figure 3.1 – A Demand Schedule and Demand Curve

Figure 3.1 A Demand Schedule and Demand Curve

As the price changes, consumers change the quantity of athletic shoes they are willing to buy. We can show this as a *demand schedule* in a table or as a *demand curve* on a graph. The table and graph both show that as the price of athletic shoes falls, the quantity demanded increases. When the price of a pair of athletic shoes is \$150, consumers buy 8 million pairs per week. When the price drops to \$125, consumers buy 9 million pairs per week. Therefore, the demand curve for athletic shoes is downward sloping.

Demand Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	8
125	9
100	10
75	11
50	12



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Figure 3.1 – Demand Curve and Ceteris Paribus (2 of 3)

- When drawing the demand curve, we assume **ceteris paribus**.
- **Ceteris paribus** (“all else equal”) condition:
 - When analyzing the relationship between two variables (such as price and quantity demanded), **other variables must be held constant**.

Figure 3.1 – Quantity Demanded and the Law of Demand (3 of 3)

- **Quantity demanded:** the amount of a good or service that a consumer is willing and able to purchase at a given price.
- **Law of demand:**
 - Holding everything else constant, when the price of a product falls, the quantity demanded of the product will **increase**.
 - When the price of a product rises, the quantity demanded of the product will **decrease**.

What Explains the Law of Demand?

- When the price of a good falls, two key effects occur:
 - **Substitution effect:** consumers substitute toward the good whose price has fallen.
 - **Income effect:** consumers have more purchasing power, which is like an increase in income.

- **Substitution effect:**

- The change in the quantity demanded of a good that results from a change in price, making the good more or less expensive **relative to other goods**, holding constant the effect of the price change on consumer purchasing power.

- **Income effect:**

- The change in the quantity demanded of a good that results from the effect of a change in the good's price on consumers' **purchasing power**, holding all other factors constant.

Everyday Example: Movie Tickets

- Suppose movie ticket prices fall from \$15 to \$10.
- **Substitution effect:**
 - Movies are cheaper relative to other entertainment (e.g., bowling, dining out), so you substitute toward movies.
- **Income effect:**
 - You effectively have more real income; your budget goes further, so you can afford more movie nights per month.

Figure 3.2 – Shifting the Demand Curve (1 of 2)

- A change in something **other than price** that affects demand causes the **entire demand curve to shift**.
- A shift to the right (D_1 to D_2) is an **increase in demand**.
- A shift to the left (D_1 to D_3) is a **decrease in demand**.

Figure 3.2 – Shifting the Demand Curve

Figure 3.2 Shifting the Demand Curve

When consumers increase the quantity of a product they want to buy at a given price, the market demand curve shifts to the right, from D_1 to D_2 . When consumers decrease the quantity of a product they want to buy at a given price, the demand curve shifts to the left from D_1 to D_3 .

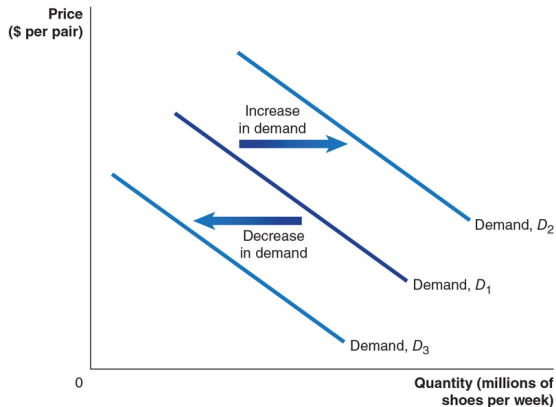


Figure 3.2 – Demand Shift vs. Quantity Demanded (2 of 2)

- As the demand curve shifts, the quantity demanded will change, **even if the price does not change**.
- The quantity demanded changes at **every possible price**.

Variables That Shift Market Demand

- Income of consumers
- Prices of related goods
- Tastes
- Population and demographics
- Expectations

Changes in Income of Consumers

- **Normal goods:** goods for which demand increases as income rises and decreases as income falls.
 - Examples: new clothes, restaurant meals, vacations.
- **Inferior goods:** goods for which demand increases as income falls and decreases as income rises.
 - Examples: second-hand clothes, instant noodles.

Effects of Changes in Income

- An increase in income will:
 - Increase the demand for new clothes (normal good), *ceteris paribus*.
 - Decrease the demand for second-hand clothes (inferior good).

Apply the Concept: Lobster as Inferior vs. Normal Good

- In some parts of the Maritimes, lobster is an **inferior good**: as incomes rise, people eat *less* lobster.
- In large cities, lobster is a **normal good**: as incomes rise, people eat *more* lobster.
- Key lesson:
 - Whether a good is normal or inferior depends on **consumer behaviour** and context, not its intrinsic characteristics.

Changes in the Price of Related Goods

- **Substitutes:** goods and services that can be used for the same purpose.
 - Examples: Big Mac and Whopper; Ford F-150 and Dodge Ram; jeans and khakis.
- **Complements:** goods and services that are used together.
 - Examples: Big Mac and McDonald's fries; hot dogs and hot dog buns.

Effects of Changes in the Price of Related Goods

- An increase in the price of a Big Mac:
 - **Increases** the demand for Whoppers (substitute).
 - **Decreases** the demand for McDonald's fries (complement).

Changes in Tastes

- If consumer tastes change, they may buy more or less of a product.
- Example:
 - If consumers become more concerned about healthy eating, they might decrease their demand for fast food.

Changes in Population and Demographics

- **Demographics:** characteristics of a population (age, race, gender, etc.).
- Increases in the **number of people** buying something will increase the amount demanded.
- Example:
 - An increase in the elderly population increases the demand for medical care.

Changes in Expectations About Future Prices

- Consumers decide not only **which** products to buy but also **when** to buy them.
- Future products are **substitutes** for current products.
- An expected **increase** in price tomorrow $\Rightarrow$ increases demand today.
- An expected **decrease** in price tomorrow $\Rightarrow$ decreases demand today.
- Example: If you find out that the price of gasoline will go up tomorrow, you increase your demand today.

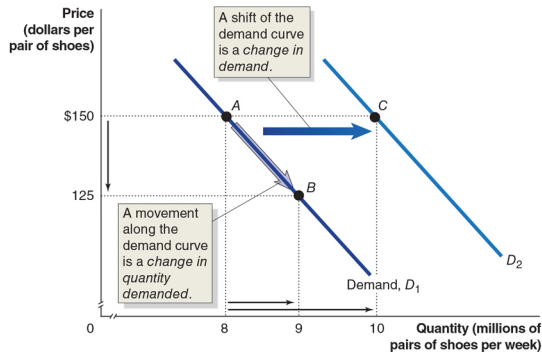
Figure 3.3 – Change in Demand vs. Change in Quantity Demanded

- A change in the **price of the product itself** causes a **movement along** the demand curve:
 - This is a **change in quantity demanded**.
- Any other change affecting demand causes the **entire demand curve to shift**:
 - This is a **change in demand**.

Figure 3.3

Figure 3.3 A Change in Demand versus a Change in Quantity Demanded

If the price of athletic shoes falls from \$150 to \$125 per pair, the result will be a movement along the demand curve from point A to point B—an increase in quantity demanded from 8 million pairs to 9 million pairs. If consumers' incomes increase, or if another factor changes that makes consumers want more athletic shoes at every price, the demand curve will shift to the right—an increase in demand. In this case, the increase in demand from D_1 to D_2 causes the quantity of athletic shoes demanded at a price of \$150 to increase from 8 million pairs at point A to 10 million pairs at point C.



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3.2 Learning Objective

Goal

Discuss the variables that influence supply.

- There are similarities and important differences between the demand and supply sides.
- Now we examine **market supply**: decisions of firms about how much of a product to provide at various prices.

Figure 3.4 – A Supply Schedule and Supply Curve (1 of 2)

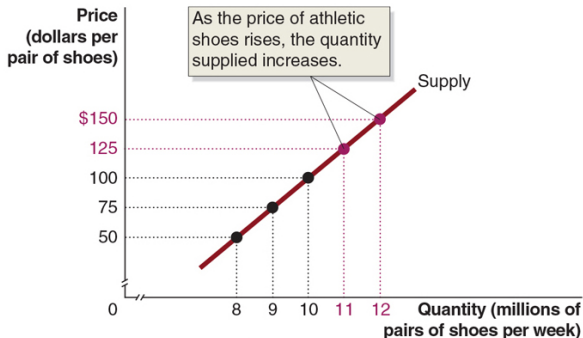
- **Supply schedule:** a table that shows the relationship between the price of a product and the quantity of the product supplied.
- **Supply curve:** a curve that shows the relationship between the price of a product and the quantity of the product supplied.

Figure 3.4 – Supply Schedule and Curve

Figure 3.4 A Supply Schedule and Supply Curve

As the price changes, Nike, Adidas, Puma, and other firms producing athletic shoes change the quantity they are willing to supply. We can show this as a *supply schedule* in a table or a *supply curve* on a graph. The supply schedule and supply curve both show that as the price of athletic shoes rises, firms will increase the quantity they supply. At a price of \$125 per pair, firms will supply 10 million pairs. At a price of \$150 per pair, firms will supply 11 million pairs.

Supply Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	12
125	11
100	10
75	9
50	8



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Figure 3.4 – Quantity Supplied and Law of Supply (2 of 2)

- **Quantity supplied:** the amount of a good or service that a firm is willing and able to supply at a given price.
- **Law of supply:**
 - Holding everything else constant, increases in price cause **increases** in quantity supplied.
 - Decreases in price cause **decreases** in quantity supplied.

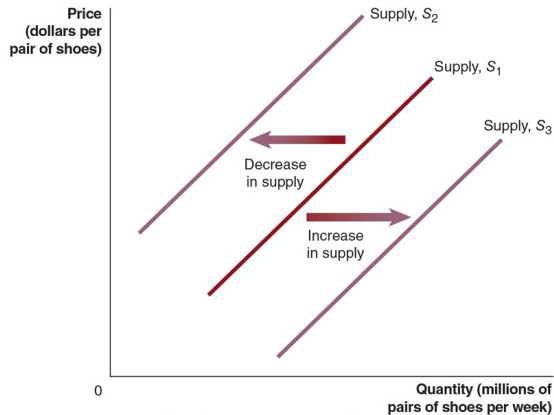
Figure 3.5 – Shifting the Supply Curve (1 of 2)

- A change in something **other than price** that affects supply causes the entire supply curve to **shift**.
- A shift to the right (S_1 to S_3) is an **increase in supply**.
- A shift to the left (S_1 to S_2) is a **decrease in supply**.

Figure 3.5 – Shifting the Supply Curve

Figure 3.5 Shifting the Supply Curve

When firms increase the quantity of a product they want to sell at a given price, the supply curve shifts to the right. The shift from S_1 to S_3 represents an *increase in supply*. When firms decrease the quantity of a product they want to sell at a given price, the supply curve shifts to the left. The shift from S_1 to S_2 represents a *decrease in supply*.



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Figure 3.5 – Supply Shift vs. Quantity Supplied (2 of 2)

- As the supply curve shifts, the quantity supplied changes **even if the price does not change**.
- The quantity supplied changes at every possible price.

Variables That Shift Market Supply

- Prices of inputs
- Technological change
- Prices of substitutes in production
- Number of firms in the market
- Expected future prices

Changes in Prices of Inputs

- **Inputs** are anything used in the production of a good or service.
 - For athletic shoes: rubber, plastic, and labour.
- An **increase** in the price of an input decreases the profitability of selling the good, causing a **decrease in supply**.
- A **decrease** in the price of an input increases profitability, causing an **increase in supply**.

- A firm may experience a positive or negative change in its ability to produce a given level of output with a given quantity of inputs.
- This is called **technological change**.
- Examples:
 - A new, more efficient way of producing shoes $\Rightarrow$ **increases** supply.
 - Government restrictions on how much workers can work $\Rightarrow$ **decreases** supply.

Prices of Substitutes and Complements in Production

- Many firms can produce and sell alternative products: **substitutes in production**.
- Example:
 - An Ontario farmer can plant either corn or soybeans.
 - If the price of soybeans rises, the farmer will plant (supply) less corn.
- Sometimes two products are necessarily produced together: **complements in production**.
- Example:
 - Cattle provide both beef and leather.
 - An increase in the price of beef encourages more cattle farming, which increases the supply of leather.

Number of Firms and Expected Future Prices

- More firms in the market $\Rightarrow$ more product available at a given price (greater supply).
- If a firm anticipates that the price of its product will be **higher** in the future, it might **decrease** its supply today to increase it in the future.

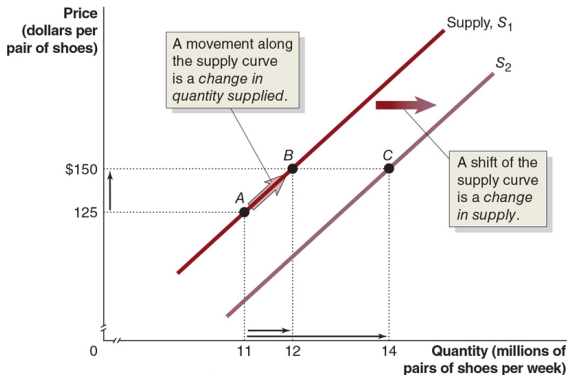
Figure 3.6 – Change in Supply vs. Change in Quantity Supplied

- A change in the **price of the product itself** causes a **movement along** the supply curve:
 - This is a **change in quantity supplied**.
- Any other change affecting supply shifts the entire supply curve:
 - This is a **change in supply**.

Figure 3.6

Figure 3.6 A Change in Supply versus a Change in Quantity Supplied

If the price of athletic shoes rises from \$125 to \$150, the result will be a movement along the supply curve from point *A* to point *B*—an increase in quantity supplied by Nike, Adidas, Reebok, and other firms from 11 million pairs per week to 12 million pairs. If the price of an input decreases, or if another factor changes that causes sellers to supply more of the product at every price, the supply curve will shift to the right—an increase in supply from S_1 to S_2 causes the quantity of athletic shoes supplied at a price of \$150 to increase from 12 million pairs at point *B* to 14 million pairs at point *C*.



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3.3 Learning Objective

Goal

Use a graph to illustrate market equilibrium.

- **Market equilibrium:** quantity demanded equals quantity supplied.
- In markets with many buyers and sellers (perfectly competitive), we call this a **competitive market equilibrium**.

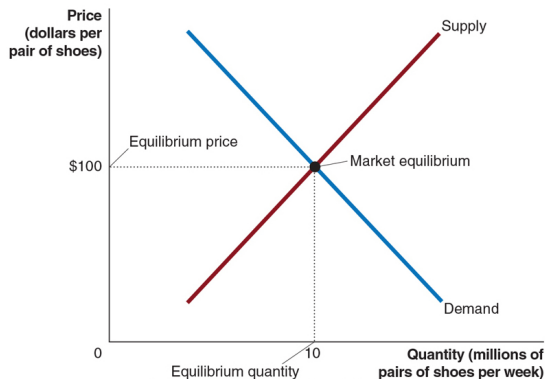
Figure 3.7 – Market Equilibrium (Athletic Shoes)

- Consider the market for athletic shoes.
- At a price of \$100:
 - Consumers want to buy 10 million pairs of shoes per week.
 - Producers want to sell 10 million pairs of shoes per week.
- The **equilibrium price** is \$100, and the **equilibrium quantity** is 10 million pairs per week.

Figure 3.7 – Market Equilibrium

Figure 3.7 Market Equilibrium

Where the demand curve crosses the supply curve determines *market equilibrium*. In this case, the demand curve for athletic shoes crosses the supply curve at a price of \$100 and a quantity of 10 million pairs. Only at this point is the quantity of athletic shoes consumers are willing to buy equal to the quantity that Nike, Adidas, New Balance, and other firms are willing to sell: The quantity demanded is equal to the quantity supplied.



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Figure 3.8 – Surpluses and Shortages (1 of 2)

- What if the price were \$125 instead?
- At \$125:
 - Consumers want to buy 9 million pairs of shoes.
 - Producers want to sell 11 million pairs of shoes.
- This gives a **surplus** of 2 million pairs: quantity supplied exceeds quantity demanded.
- Prediction: sellers will compete among themselves, driving the price **down** toward equilibrium.

Figure 3.8 – Surplus

Figure 3.8 The Effect of Surpluses and Shortages on the Market

When the market price is above equilibrium, there will be a *surplus*. A price of \$125 for athletic shoes results in 11 million pairs being supplied but only 9 million pairs being demanded, or a surplus of 2 million pairs. As Nike, Adidas, New Balance, and other firms cut the price to dispose of the surplus, the price will fall to the equilibrium price of \$100. When the market price is below equilibrium, there will be a *shortage*. A price of \$50 results in 12 million pairs being demanded but only 8 million being supplied, or a shortage of 4 million. As firms find that consumers who are unable to find athletic shoes available for sale are willing to pay higher prices to get them, the price will rise to the equilibrium price of \$100.

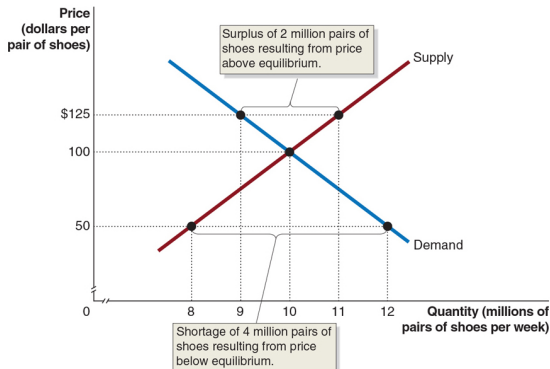


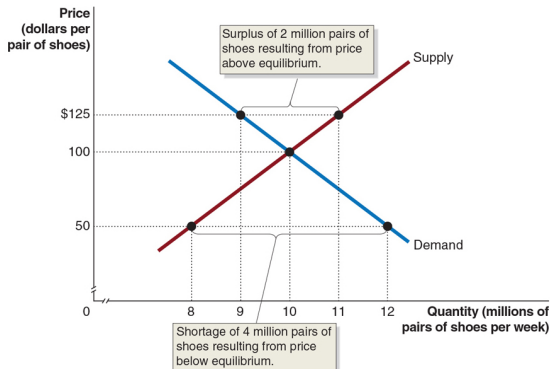
Figure 3.8 – Surpluses and Shortages (2 of 2)

- What if the price were \$50?
- At \$50:
 - Consumers want to buy 12 million pairs of shoes.
 - Producers want to sell 8 million pairs.
- This gives a **shortage** of 4 million pairs: quantity demanded exceeds quantity supplied.
- Prediction: sellers realize they can increase price and still sell as many pairs $\Rightarrow$ price will **rise** toward equilibrium.

Figure 3.8 – Shortage

Figure 3.8 The Effect of Surpluses and Shortages on the Market

When the market price is above equilibrium, there will be a *surplus*. A price of \$125 for athletic shoes results in 11 million pairs being supplied but only 9 million pairs being demanded, or a surplus of 2 million pairs. As Nike, Adidas, New Balance, and other firms cut the price to dispose of the surplus, the price will fall to the equilibrium price of \$100. When the market price is below equilibrium, there will be a *shortage*. A price of \$50 results in 12 million pairs being demanded but only 8 million being supplied, or a shortage of 4 million. As firms find that consumers who are unable to find athletic shoes available for sale are willing to pay higher prices to get them, the price will rise to the equilibrium price of \$100.



Demand and Supply Both Count

- Price is determined by the **interaction** of buyers and sellers.
- In a competitive market, neither group alone can dictate the price.
- However, **changes in demand and/or supply** will affect the equilibrium price and quantity traded.

3.4 Learning Objective

Goal

Use demand and supply graphs to predict changes in prices and quantities.

- We typically know the actual price and quantity, but do *not* observe the whole curves.
- The power of the model is in predicting **directional changes** in price and quantity when curves shift.

Figure 3.9 – Effect of an Increase in Supply (1 of 2)

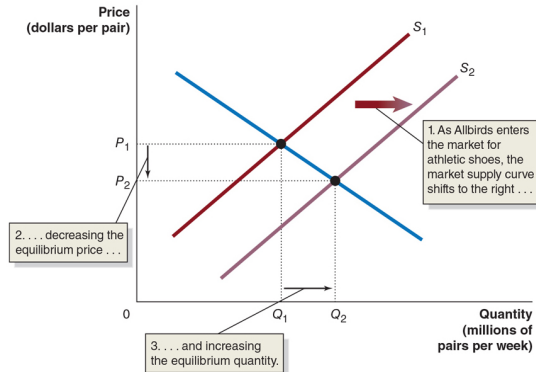
- Market for athletic shoes when a new shoemaker enters the market.
- At any given price, more shoes are supplied:
 - Supply shifts right from S_1 to S_2 .
- **Result:**
 - Equilibrium price **falls**.
 - Equilibrium quantity **rises**.

Figure 3.9 – Increase in Supply

Figure 3.9 The Effect of an Increase in Supply on Equilibrium

If a firm enters a market, as Allbirds did with its Trail Runner SWT, the equilibrium price will fall and the equilibrium quantity will rise:

1. As Allbirds enters the market for athletic shoes, a larger quantity of athletic shoes will be supplied at every price, so the market supply curve shifts to the right, from S_1 to S_2 , which causes a surplus of athletic shoes at the original price, P_1 .
2. The equilibrium price falls from P_1 to P_2 .
3. The equilibrium quantity rises from Q_1 to Q_2 .



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Figure 3.9 – How Much Does Price Fall? (2 of 2)

- By how much will price fall?
- By how much will quantity rise?
- We **cannot** say without knowing more (the exact shapes and positions of demand and supply).
- For now, we only predict *directions*: price falls, quantity rises.

Figure 3.10 – Effect of an Increase in Demand

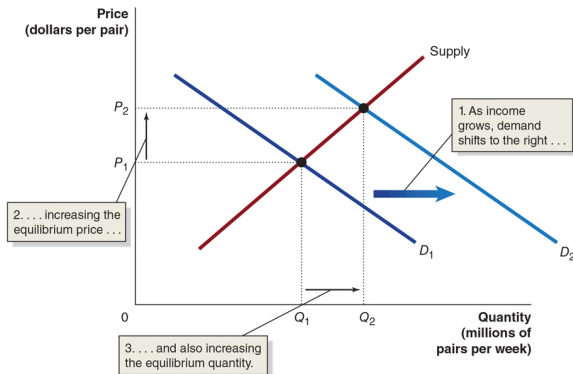
- Suppose incomes increase.
- Athletic shoes are a **normal good**, so as income rises, demand shifts to the right.
- **Result:**
 - Equilibrium price **rises**.
 - Equilibrium quantity **rises**.

Figure 3.10 – Increase in Demand

Figure 3.10 The Effect of an Increase in Demand on Equilibrium

Increases in income will cause equilibrium price and quantity to rise:

1. Because athletic shoes are a normal good, as income grows, the quantity demanded increases at every price and the market demand curve shifts to the right, from D_1 to D_2 , which causes a shortage of athletic shoes at the original price, P_1 .
2. The equilibrium price rises from P_1 to P_2 .
3. The equilibrium quantity rises from Q_1 to Q_2 .



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Table 3.3 – How Shifts in Demand and Supply Affect Equilibrium P and Q

(1 of 2)

	Equilibrium Price P	Equilibrium Quantity Q
Increase in demand, supply unchanged	Rises	Rises
Decrease in demand, supply unchanged	Falls	Falls
Increase in supply, demand unchanged	Falls	Rises
Decrease in supply, demand unchanged	Rises	Falls

Table 3.3 – How Shifts in Demand and Supply Affect P and Q (2 of 2)

	Equilibrium Price P	Equilibrium Quantity Q
Increase in supply, demand unchanged	Falls	Rises
Increase in demand, decrease in supply	Rises	?
Decrease in demand, increase in supply	Falls	?
Increase in demand, increase in supply	?	Rises
Decrease in demand, decrease in supply	?	Falls

- The highlighted row (in blue text) is the example we studied: **increase in supply**.
- “?” means the effect is **ambiguous** without more information on the relative size of shifts.

Figure 3.11 – Shifts in Demand and Supply Over Time (1 of 3)

- Over time, it is likely that **both** demand and supply will change.
- Example:
 - New firms enter $\Rightarrow$ supply shifts right.
 - Incomes rise $\Rightarrow$ demand shifts right.

Figure 3.11 – Both Demand and Supply Shift

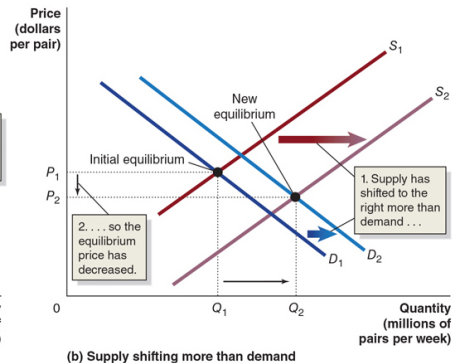
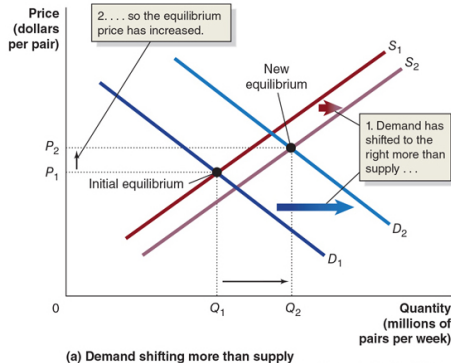
Figure 3.11 Shifts in Demand and Supply

Whether the price of a product rises or falls over time depends on whether demand shifts to the right more than supply. In panel (a), demand shifts to the right more than supply, and the equilibrium price rises:

1. Demand shifts to the right more than supply.
2. The equilibrium price rises from P_1 to P_2 .

In panel (b), supply shifts to the right more than demand, and the equilibrium price falls:

3. Supply shifts to the right more than demand.
4. The equilibrium price falls from P_1 to P_2 .



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Figure 3.11 – Case 1: Demand Shifts More than Supply (2 of 3)

- When demand shifts more than supply:
 - Equilibrium quantity **rises**.
 - Equilibrium price also **rises**.
- This panel shows a larger rightward shift of demand than of supply.

Figure 3.11 – Case 2: Supply Shifts More than Demand (3 of 3)

- When supply shifts more than demand:
 - Equilibrium quantity **rises**.
 - Equilibrium price **falls**.
- Without knowing the relative size of the changes, the effect on price is ambiguous.
- It is possible, but unlikely, that equilibrium price remains unchanged.

Shifts of Curves vs. Movements Along Curves

- Suppose an **increase in supply** occurs.
- We know:
 - Equilibrium quantity will increase.
 - Equilibrium price will decrease.
- It is tempting to say “the decrease in price causes an increase in demand”, but this is incorrect.
- The decrease in price causes a **movement along the demand curve**, not a shift of the curve.
- The demand curve already describes how much consumers want to buy at **any given price**.

- Demand:
 - Follows the law of demand; explained by substitution and income effects.
 - Shifts when income, prices of related goods, tastes, population, or expectations change.
- Supply:
 - Follows the law of supply.
 - Shifts when input prices, technology, related goods in production, number of firms, or expected future prices change.
- Market equilibrium:
 - Occurs where quantity demanded equals quantity supplied.
 - Surpluses cause price to fall; shortages cause price to rise.
- The demand and supply model predicts **directional** changes in equilibrium price and quantity when curves shift.

Thank you!